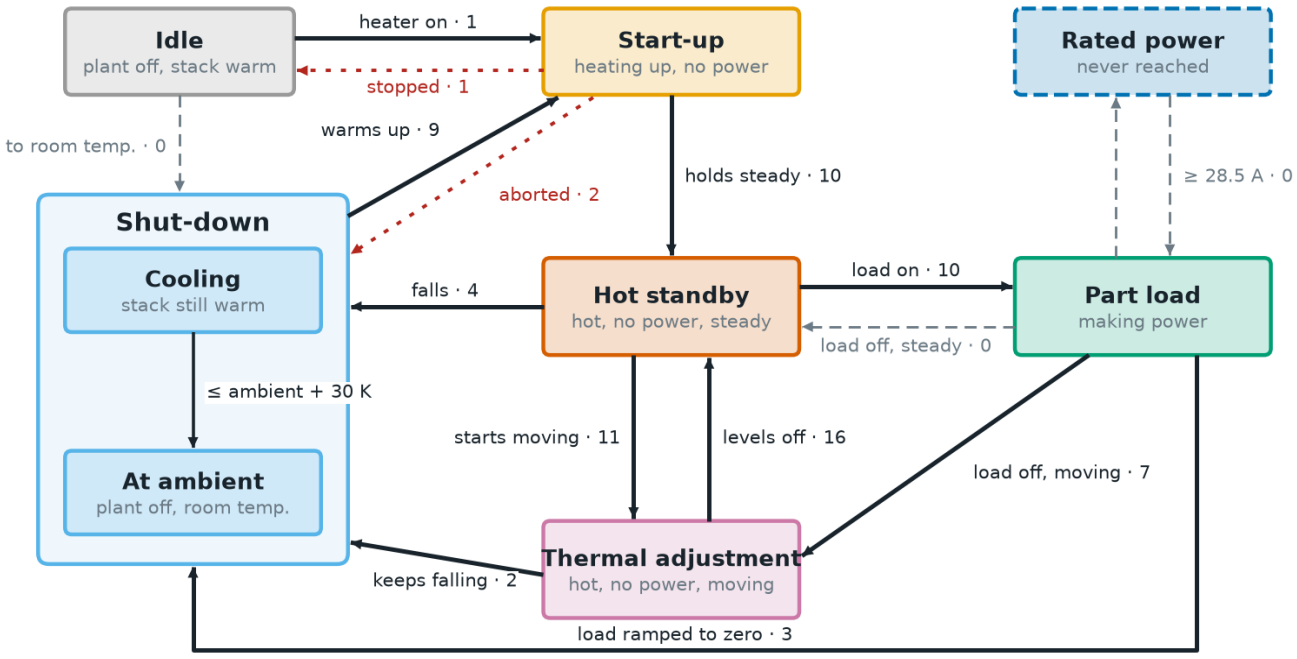
Seven stages, one fixed order: cold stacks only start, and power only comes from a steady hot stack

Contents: 1 The stage map · 2 Each stage, in and out · 3 How one minute is labelled · 4 Who may follow whom

Every minute of the 39 plant days gets one stage. The stages may only follow each other along the arrows below. Each arrow has a plain condition, and the number on it says how often it really happened.

Why it matters: the fake-data generator builds its timelines from this map. A missing arrow can never be produced; a wrong arrow spreads into every synthetic day.



Solid = allowed and seen (count = times in 39 days). Dashed grey = allowed, not seen yet. Red dotted = seen but outside the rules (flagged). Rated power would connect like Part load; it never happened.

The four rules of order

Start only from cold. A Start-up can only follow a Shut-down (or Idle). A hot stack is never “started”.

Power only from Hot standby. Load always begins from a steady Hot standby; the natural-gas minutes just before load are Hot standby “pre-load”.

A hot stack that moves is adjusted. No load and the temperature rising or falling is Thermal adjustment. It returns to Hot standby or slides into Shut-down.

Shut-down ends only in Start-up. Inside it, Cooling comes before At ambient.

If you remember one thing: Shut-down → Start-up → Hot standby is the only way up; from Hot standby the plant goes to load, adjusts its temperature, or shuts down.

2 · EACH STAGE, IN AND OUT

Every stage has a test to be in it, a test to leave it, and a short list of stages it may hand over to

Room temperature is not measured directly; the fuel-line probes stand in, so “within 30 K of room temperature” means about 40–48 °C.

Idle	2 episodes · 21.0 h
Comes from	(start of data)
Goes to	Start-up, Shut-down
In it when	Plant off (no current, heater off, air off), stack more than 30 K above room temperature, and no Shut-down before it.
Ends when	Heater comes on (→ Start-up) or the stack reaches room temperature (→ Shut-down).
Gas / air	Plant off
Inside it	Natural cooling (heater off, no air) 19.3 h, Heat-up stopped – settling 1.6 h
Note	Not a real plant state: only the first file, 2025-10-30.

Start-up	14 episodes · 148.9 h
Comes from	Shut-down, Idle
Goes to	Hot standby
In it when	Heater on with air flowing below 550 °C, or temperature rising more than 10 °C per hour, and the stack has not been hot since the last Shut-down.
Ends when	The temperature holds steady → Hot standby.
Gas / air	Air only first, forming gas (N2 + H2) from about 190–240 °C.
Inside it	Heat-up (heater on) 117.5 h, Heat-up interrupted (plant off, 03-17/18) 11.0 h, Final heat-up to Hot standby 6.8 h, Slow heat-up 4.8 h, Air flow start (purge) 3.4 h, Short pause / dip / hold 5.3 h
Note	Heats at about 50 °C per hour; a stop of up to 3 h stays inside it.

Hot standby	26 episodes · 297.3 h
Comes from	Start-up, Thermal adjustment, Part load, Rated power
Goes to	Part load, Rated power, Thermal adjustment, Shut-down
In it when	No load, temperature constant (under 5 °C per hour), at 550 °C or more, or 450–550 °C held for 6 hours or more.
Ends when	Load comes on, the temperature starts moving, or a cool-down begins.
Gas / air	Forming gas; natural gas only just before load.
Inside it	Low-temperature hold (~510 °C) 191.4 h, Temperature hold (575–610 °C) 103.9 h, Pre-load (natural gas on, load next) 1.9 h, Fuel switch-over 0.1 h
Note	The only door into load.

Thermal adjustment	18 episodes · 45.8 h
Comes from	Hot standby, Part load, Rated power
Goes to	Hot standby, Shut-down
In it when	No load, stack hotter than 450 °C, temperature moving by 5 °C per hour or more, after the stack has been hot.
Ends when	Levels off (→ Hot standby) or keeps falling into the cool-down (→ Shut-down).
Gas / air	Forming gas.
Inside it	Cool-back 17.5 h, Re-heat to operating temperature 13.8 h, Re-heat 10.6 h, Fuel switch-over 1.5 h, Re-heat to hold 1.4 h, Dip during hold 1.0 h
Note	If it runs straight into load, its last minutes become Hot standby pre-load.

Part load	10 episodes · 4.5 h
Comes from	Hot standby, Rated power
Goes to	Thermal adjustment, Shut-down, Hot standby, Rated power
In it when	Current of 1 A or more per string for at least 3 minutes, below 28.5 A.
Ends when	Load goes off.
Gas / air	Always natural gas.
Inside it	Load ramp-up 4.0 h, Steady load 0.3 h, Load ramp-down 0.2 h
Note	Highest seen: 16.2 A, 33.6 kW. 8 of 10 runs ended in a safety trip.

Rated power	0 episodes · 0.0 h
Comes from	Hot standby, Part load
Goes to	Part load, Hot standby, Thermal adjustment, Shut-down
In it when	Current of 28.5 A per string or more (95 % of the rated 30 A).
Ends when	Current drops or load goes off.
Gas / air	Natural gas.
Inside it	no minutes in the data
Note	Never reached in the 39 days.

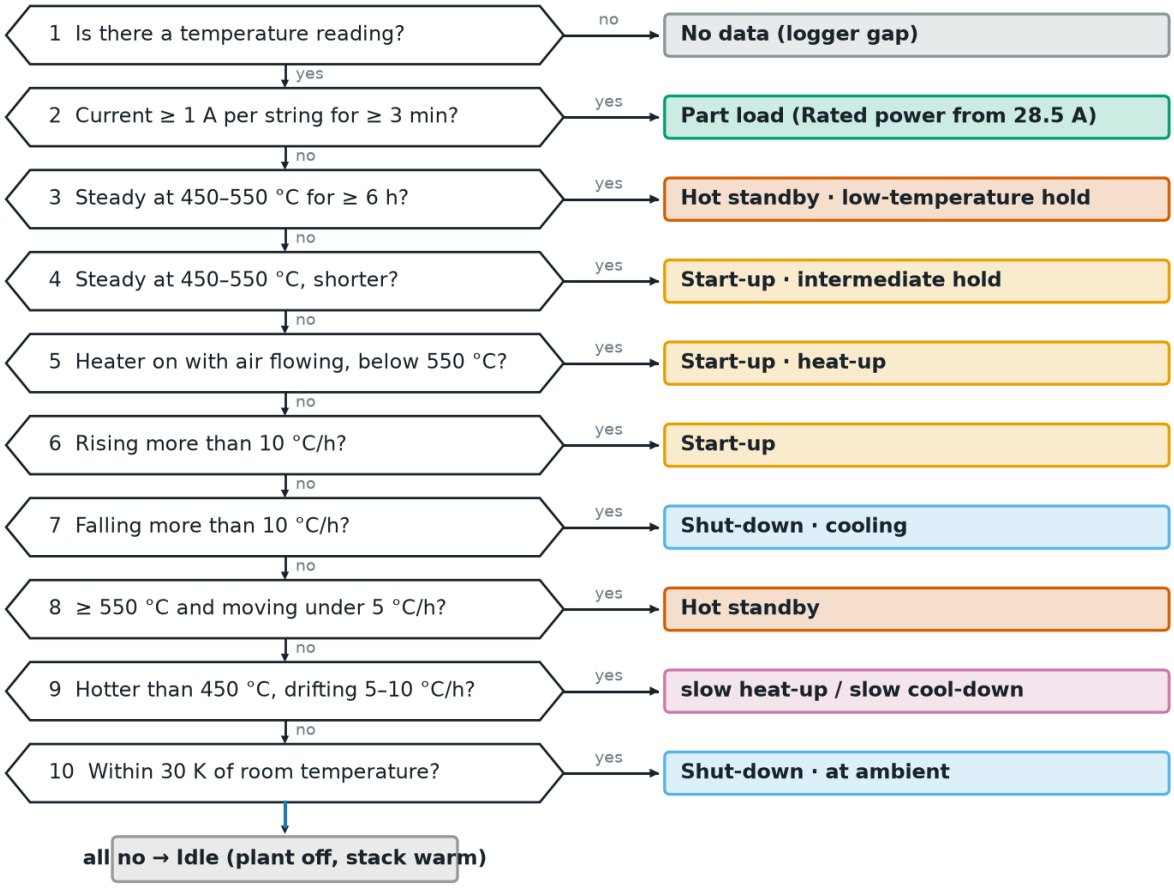
Shut-down	19 episodes · 396.1 h
Comes from	Hot standby, Part load, Thermal adjustment, Rated power, Idle
Goes to	Start-up
In it when	Cooling: load ramped to zero, temperature falling more than 10 °C per hour, cooling with air on, or natural cooling after the air goes off. At ambient: plant off and stack within 30 K of room temperature.
Ends when	Only a new Start-up ends it. Inside, Cooling always comes before At ambient.
Gas / air	Forming gas down to about 210–290 °C, then air only, then plant off.
Inside it	Cool-down (air on) 248.3 h, Plant off at ambient 77.6 h, Natural cooling (air off) 52.8 h, Heater on without air (false start) 10.6 h, Slow cool-down 5.0 h, Other (trim, ramp-down) 1.9 h
Note	Two parts: Cooling (308 h) and At ambient (88 h).

If you remember one thing: Hot standby is the hub: it is the only stage that can hand over to load, and almost every path passes through it.

3 · HOW ONE MINUTE IS LABELLED

Ten questions in a fixed order give each minute a first label; a second look at the whole day fixes what one minute cannot see

The first “yes” wins, so any minute with power flowing is load, whatever the temperature does. Steady = changing less than 5 °C per hour; rising or falling = more than 10 °C per hour.



Second look at the whole day

What the first pass sees	What it becomes
Hot stack, no load, temperature moving	Thermal adjustment (re-heat or cool-back)
Idle right after a Shut-down	Shut-down · natural cooling (air off)
Natural-gas minutes running straight into load	Hot standby · pre-load (15 min if no gas)
Natural gas on, no load following	operation “Fuel switch-over” inside its stage
Heater on but blowers off (2026-02-06)	Shut-down · at ambient, “false start”
Cool-back that runs on into the cool-down	start of that Shut-down
Heater pause up to 3 h inside a heat-up	stays Start-up
Overnight stop 2026-03-17/18 between two heat-up parts	stays Start-up (“heat-up interrupted”)
Air on, cooling just before a cold heat-up	Start-up · air flow start (purge)
Any block shorter than 15 min	joins its neighbour (load minutes never merge)

If you remember one thing: Current first, then the long hot hold, then the heater, then the direction of the temperature; context rules clean up afterwards.

Of the 42 possible stage changes, 18 are allowed; 10 of them actually happened, 73 times

Read a row for where a stage may go, a column for where it may come from. Teal = allowed and seen, white 0 = allowed but not seen, red = broke the rules (flagged), dot = forbidden.

from \ to	Idle	Start-up	Hot standby	Thermal adj.	Part load	Rated power	Shut-down
Idle		1	·	·	·	·	0
Start-up	1		10	·	·	·	2
Hot standby	·	·		11	10	0	4
Thermal adjustment	·	·	16		·	·	2
Part load	·	·	0	7		0	3
Rated power	·	·	0	0	0		0
Shut-down	·	9	·	·	·	·	

Why the dots are forbidden

Into	Reason
Start-up	A hot stack is never “started”. Warming a hot stack is Thermal adjustment. Start-up only follows Shut-down (or Idle).
Part load	Power only after a steady Hot standby. Every load begins from Hot standby (pre-load).
Hot standby	A stack that was cooled or is still heating up first has to finish its Start-up.
Thermal adjustment	Only a stack that has already been hot (Hot standby or load) can be adjusted.
Idle	Idle only exists when no Shut-down comes before it, so nothing on the map leads into it.
Shut-down	Start-up cannot be abandoned into Shut-down by design (the 2 real cases are flagged).
Rated power	Rated power, like Part load, only follows Hot standby.

Every allowed or seen change

From → to	Condition	Times
Shut-down → Start-up	Heater and air come on and the stack warms by more than 10 °C per hour.	9
Idle → Start-up	Heater comes on while the stack is still warm (first file, 2025-10-30).	1
Idle → Shut-down	The warm stack cools to within 30 K of room temperature.	0 (not seen)
Start-up → Hot standby	Temperature stops moving (under 5 °C per hour) at 550 °C or more, or holds steady at 450–550 °C for 6 hours or more.	10
Hot standby → Part load	Current reaches 1 A per string for at least 3 minutes. The natural-gas minutes just before it count as Hot standby “pre-load”.	10
Hot standby → Thermal adjustment	Still no load, but the temperature starts moving by 5 °C per hour or more (re-heat or cool-back).	11
Thermal adjustment → Hot standby	The temperature levels off again (under 5 °C per hour).	16
Part load → Thermal adjustment	Load goes off and the hot stack's temperature drifts.	7
Hot standby → Shut-down	Temperature falls by more than 10 °C per hour and keeps falling into the cool-down.	4
Part load → Shut-down	Load is ramped to zero and the cool-down follows straight away.	3
Thermal adjustment → Shut-down	A cool-back does not level off and runs on into the cool-down.	2

From → to	Condition	Times
Part load → Hot standby	Load goes off and the temperature stays steady.	0 (not seen)
Hot standby → Rated power	Load starts straight at 28.5 A per string or more.	0 (not seen)
Part load → Rated power	Current rises past 28.5 A per string (95 % of the rated 30 A).	0 (not seen)
Rated power → Part load	Current drops below 28.5 A per string.	0 (not seen)
Rated power → Hot standby	Load goes off and the temperature stays steady.	0 (not seen)
Rated power → Thermal adjustment	Load goes off and the temperature drifts.	0 (not seen)
Rated power → Shut-down	Load ramped to zero and cool-down follows.	0 (not seen)
Start-up → Shut-down	A heat-up abandoned before Hot standby and cooled again (2025-11-13, 2026-01-30).	2 (flagged)
Start-up → Idle	A heat-up stopped at about 215 °C, plant switched off (2025-10-30).	1 (flagged)

Still open: should an abandoned heat-up (Start-up → Shut-down) become an allowed, rare arrow? Today the 2 real cases stay flagged.

If you remember one thing: Each row has very few doors: Start-up has one way out, Shut-down has one way out, and only Hot standby leads to power.